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Product Requirements Document (PRD) v2.0 – Week 13 Update

Hilo Eco-tourism Rental

Term Project - Full Stack Website - Spring 26

OVERVIEW

Problem statement: A website app to service (marketing, booking, contact) for a vacation rental property in Hilo, Hawaii.

Property/Audience: The target market is eco-tourism due to the close proximity to Kilauea, Mauna Loa, Mauna Kea, and beaches along the Big Island's windward coastline. Hilo was chosen (rather than a location closer to Kilauea) because of the amenities found in the larger town, hopefully opening up to a broader range of customers.

Website Requirements: The site will consist of three/four pages:
1) a) a marketing page that includes a hero section, b) a dashboard for displaying charts/stats (from stored records) and current weather (from an API), c) amenities of the property listing, d) a way to get to the booking page.
2) a booking page/modal where the property can be reserved for an amount of time – with confirmation.
3) a) an About/Contact page for linking further communications. b) ethos for the owner/renter
4) and an Admin page for updating content.

UX/ CSS Design – All pages will include responsive design (grid/flexbox, UX feedback), be accessible (proper headers, alt-text, >4.5 contrast, mobile-friendly at 375px), and keep a consistent design to invoke similarity.

Architecture – The app will use the following:
1) React for rendering HTML/CSS in a dynamic way client-side,
2) a JavaScript Node.js environment running Express.js for handling backend routes/ REST API needs,
3) MongoDB/Mongoose.js for storing data (DBEDT records, property info, reviews, bookings) and data modeling,
4) Chart.js/D3.js for data visualization,
5) authentication using passport.js to manage admin page credentials.

This will be deployed on the internet, so security should be a concern throughout (no hardcoded/public secrets, input sanitization, HTTP security, CORS security, and password/login protections at the minimum).

Dashboard Design / Widgets – will consist of 4 parts:

1) Graph - DBEDT arrivals (by month)

=> for ??? (why would a vacation rental customer care?)

2) Pie - DBEDT visitors by origin (countries)

=> for ??? (inclusivity for visitor majority, might ostracize visitor minorities though)

3) Bar - Kilauea's eruptive frequency by week/month

=> for visitor to take predictive glance at eruptive activity

4) Weather - current weather in Hilo –

=> openweather api for Hilo's location

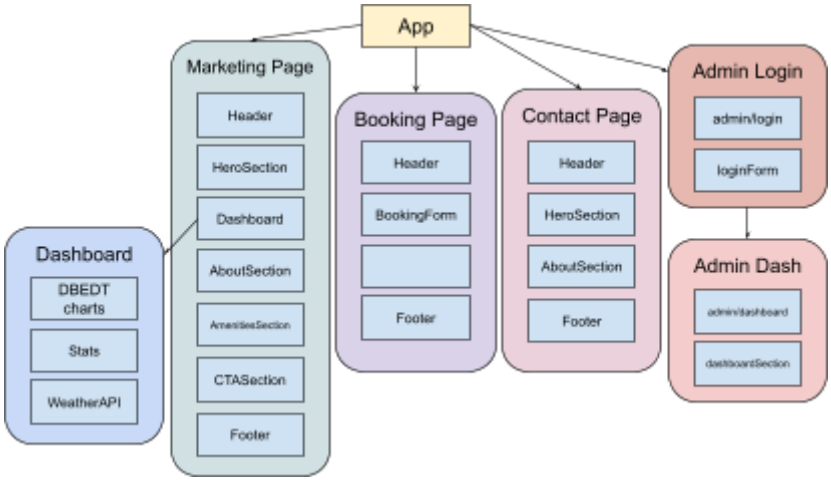
fallback => historical avg of current month

User Personas	John Doe Visitor, looking housing options, to see Kilauea	Jane Who Already booked, confirming reservation, planning trip	Operator adding/ updating content, fit into busy schedule
Information/ Functionality/ Wants	- good first impression - easy to read info - simple user flow - local recommendations (activities, food) - wifi? home rules? parking? costs? bedrooms?	- easy to find booking info - see status, add to calendar - what amenities are with property - 'springboard' to find things to do nearby - how to get to things nearby?	+ easy to use UI + easy to use login (reset password?) + see what is currently posted + age of information (since last update)
Considerations / Goals/ Needs	- reaching out for help? - why this place over somewhere else? - disability accommodations? - hotel level quality at a discount	- know who the user is - update /cancel booking? - further links to amenities if possible	+ update content — add/remove property/packages — update property info + see backend status — is db running, console errors + see website stats — # of bookings and when — page visits, etc.
Possible frustrations	- is the owner trustworthy? - Bad UX? Too easy? - others' reviews?	- dont know password - avoid information overload	+ no web development experience + avoid overly complicated + dont know password

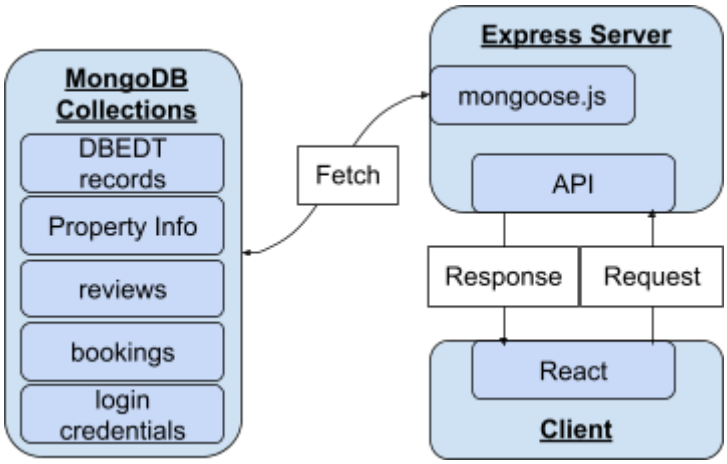
Component Requirements

<table>
<tr><td>Marketing Page</td><td>
<table>
<tr><td>Header</td><td>HeroSection</td><td>Dashboard</td><td>About Section</td><td>Amenities Section</td><td>CTASection</td><td>Footer</td></tr>
<tr><td>– logo is anchor to home page
– links to other pages</td><td>– image of house/kilauea
– “address” title
– “vacation rental” subtitle</td><td>– DBEDT charts
+ (arrivals [month/year]
+ visitor origin (pie)
+ other stats
– current weather (via api)</td><td>– property description
+ geo-tourism
– property location</td><td>– list of amenities</td><td>– [Book Now] button
– Contact Information</td><td>– links to other pages
– copyright, year</td></tr>
</table>
</td></tr>
<tr><td>Booking Page/ Modal</td><td></td></tr>
<tr><td>About Page</td><td>
<table>
<tr><td>Website Header</td><td>About HeroSection</td><td>AboutSection</td><td>Website Footer</td></tr>
<tr><td>– logo is anchor to home page
– links to other pages</td><td>– “About/Contact” title
– different image</td><td>– owner image
– owner background
– owner contact</td><td>– links to other pages
– copyright, year</td></tr>
</table>
</td></tr>
<tr><td>Admin</td><td>
<table>
<tr><td>Admin/login</td><td>Admin/dashboard</td></tr>
<tr><td>– login page
(cookies for storing credentials?
7 day grace?)</td><td>– update property/package information
(options = backend status, site stats, is info old?)</td></tr>
</table>
</td></tr>
</table>

App -> child diagram



Data Flow



APIs that are exposed:

/properties	GET /properties? – (returns JSON list of filterable properties) GET /properties:id – (returns JSON of info of one property by its MongoDB id) POST properties/:id/reviews – (add review to database)
/admin	GET login page GET/POST login try, POST login credentials? GET dashboard page POST updates to properties, post add/remove property, post add/remove package

Acceptance Criteria

1) Given User A has just landed on the marketing page, when user A wants to make a booking, then user A can successfully make a booking in 4 clicks (Make a booking, start date, end date, confirm)
2) Given User B is confused on **any** page, when wanting to ask for help/info, user B can access contact information with a link on that page.
3) Given User C has just landed on **any** page, when the page loads it takes no longer than 500ms for the page to load correctly.
4) Given User E has made a booking, when they want to see their information, they can do so in 2 clicks (see my information, verify its me)

v2.0 Scope Changes –

- Additions / Changes highlighted in yellow
- Compressed some sentence information to size
- Moved a lot of information into tables/charts for easier readability
- No longer considering foreign visitors for website accommodation, leave translation to the browser, leave marketing to a good design rather than pandering
- Added user persona for someone who is already booked / planning a trip
- Removed competence acceptance criteria since its alluded to in others

v2.0 Considerations / Future Plans

- The DBEDT charts don't currently serve a purpose in getting a user from login to booking (other than flashy ethos)
- UseState uses: filter amenities by if they are apart of the Property, in Hilo, or by Kilauea
- Add wireframe (low fidelity, high fidelity) to PRD
- Reviews (/properties/:id/reviews) needs a use case, thoughts about where to put it and why
- About Page/Modal could be a part of the marketing page only, no need for a separate page?
- Booking Page / Modal, proper setup, book by date, store information, check clashes
- Focus is only one property at the moment, though make tileable incase adding another

– making a bar/frequency chart of eruptive activity is a cool idea. Do chart by month if by week is too sparse. Look for data sources/ relevant api's?

– a google maps module that shows the location of the property -> links to opening a new page that opens google maps => convenience?